

Correction Notice — Revision 4

*Beyond Sturmian Words in the $3x+1$ Periodicity Conjecture:
A Periodic-Approximation and Arithmetic-Height Bridge*

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Summary. No mathematical statement of Revision 3 is altered or withdrawn. Revision 4 corrects two statements of fact about the deposit itself, one framing error in the open-problem list, and adds explicit numerical constants where Revision 3 asserted only effectivity. Revision 3's theorems — in particular the critical-density results for quadratic rotations at rational and at algebraic offsets — stand exactly as stated, with the same proofs.

E1. Revision 3 is deposited. Revision 3's "Provenance and Verification" section states: "At the time of writing Revision 3 is a draft and has not been deposited." It was deposited on 30 September 2026 as Zenodo version DOI [10.5281/zenodo.23068563](https://doi.org/10.5281/zenodo.23068563). Revision 4 replaces that sentence with the full deposit table.

E2. The "draft" labelling is removed. Revision 3 was deposited under a filename containing DRAFT. Deposited files cannot be renamed in place; Revision 4 is issued as `note_rev4.pdf` and nothing in it is labelled a draft.

E3. Open 4's framing is corrected — the only substantive item. Revision 3 labelled "Open 4 — unbounded partial quotients" *the live frontier*. That conflates two different things, and repeats for the unbounded-type case precisely the error Revision 3 itself corrected for the bounded-type case.

Every complementary symmetric Rote sequence has ones-density exactly $1/2$ *whatever* the slope's partial quotients: the density- $1/2$ fact concerns the doubled root of the odd-weight transfer and is independent of the continued fraction. Since $1/2 < \beta = \ln 2 / \ln 3$, Monks–Yazinski, *The autoconjugacy of the $3x + 1$ function*, Discrete Math. **275** (2004) 219–236, Theorem 2.7(b) already gives $\Phi(v) \notin \mathbb{Q}$ for every CS Rote sequence over *every* irrational slope, bounded type or not.

What is unfinished at unbounded partial quotients is therefore *the note's own independent bridge*, whose odd-weight branch has margin exactly 0 at an intercept with $\text{ice}(\omega) = 2$ — not the irrationality conclusion, which is available. Open 4 asks for a second proof of an available statement by a mechanism that presently cannot supply one. It remains worth answering, because that mechanism is what yields effective height bounds, but it is not a frontier of knowledge about irrationality. The frontier is lower ones-density $\geq \beta$, where Revision 3's new theorems operate. Revision 4 rewrites the Open 4 entry and the corresponding roadmap item, and downgrades Revision 2's judgement that Open 4(i) was "the single highest-value target".

E4. Explicit constants added. Revision 3 said the asymptotic argument was "effective" without supplying numbers. Revision 4 adds a section deriving them for the named example $\alpha = (\sqrt{5}-1)/2$, $\rho = 1/7$:

$$S(W) \geq (L - \ell) - C \log_2 \ell - C_0, \quad C = 24.207846, \quad C_0 = 35.869175,$$

from Kuipers–Niederreiter's discrepancy bound read directly at its source, and an explicit $f(M)$ with $H \geq 2^{f(M)}$ for every integer $M \geq 2$, positive for $M \geq 1.78337 \cdot 10^{18}$. In the course of that derivation one looseness in Revision 3 was identified and is recorded: Revision 3 set $\log B = \log H(A_m) + 2$ without distinguishing the absolute logarithmic Weil height appearing in the hypothesis of the

logarithmic-form estimate from the classical height being bounded. Revision 3's choice is *larger*, hence admissible — the hypotheses on B are lower bounds and the conclusion decreases in B — but wasteful; the sharper conversion $h(1 : A_m : -1) \leq \frac{1}{2} \log H(A_m) + 0.275$ is used in Revision 4. **No conclusion changes.**

The explicit threshold $M \approx 1.8 \cdot 10^{18}$ is beyond computation and is not a computational claim. It does not compete with, and does not improve, the exact finite certificates $H \geq 2^{462}, \dots, 2^{15998}$, which come from integer comparisons at witnesses of length at most 6765 and use no analytic input.

On the deposit history. The concept record 10.5281/zenodo.23045140 carries four versions. One of them, 10.5281/zenodo.23068326, contains the same file as Revision 2 byte for byte (md5:b61d76f6a3a0d23ec6d148f1aeea658b) under a filename suggesting a later build. It is noted here so that a reader arriving at it knows what it holds; it is part of the record's history and is left as it stands. **Cite Revision 2** as 10.5281/zenodo.23061396.

Acknowledgement and dependency, unchanged. López-Stoll's work on density and the $3x + 1$ conjugacy map helped guide this programme toward the critical density $\beta = \ln 2 / \ln 3$. We acknowledge that influence. The results presented here are established through the independent arguments given in the note and do not require their proposed upper-density exclusion. We take no position on their Theorem 1: we have neither established nor disproved it, and no step of any theorem in the note invokes it.

Standing scope. The note proves nothing about divergent orbits of positive integers, nothing about the Collatz conjecture, and nothing about the $3x + 1$ Periodicity Conjecture in general. Novelty of the critical-density results is formally unresolved. Neither the note nor the foundation paper it builds on (10.5281/zenodo.23019799, by the same author) has been refereed.